

Internet Access and Female Economic Empowerment: A Cross-National Nonparametric Analysis

ISyE 6404 — Nonparametric Data Analysis

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Spring 2026

Data Source: World Bank Gender Data Portal
<https://genderdata.worldbank.org/en/indicators>

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1 Introduction

Digital connectivity has become a cornerstone of modern economic participation. While the economic benefits of internet usage have been broadly explored, the relationship between national-level internet access and women’s labor market participation and advancement remains understudied [10]. This study investigates whether national internet usage rates are associated with two indicators of female economic empowerment: overall employment rates and representation in middle and senior management roles.

Understanding these relationships is relevant for development researchers, policymakers, and international organizations seeking to promote gender equity through infrastructure investment. Our analysis draws on cross-national data from the World Bank Gender Data Portal for the year 2022, the year with the most complete data across all three indicators.

We pose two primary research questions:

1. Is there a significant difference in the median female employment rate between countries with high versus low national internet usage?
2. Is there a significant difference in the median female share of middle and senior management positions between countries with high versus low national internet usage?

Because our data violates normality assumptions, we employ Mann-Whitney U tests, a nonparametric method well-suited for comparing two independent groups on a continuous outcome without assuming normality or equal variances. A Bonferroni correction is applied to control the family-wise error rate ($\alpha = 0.025$ per test).

2 Data Description

2.1 Data Sources

Three datasets were obtained from the World Bank Gender Data Portal:

1. **Individuals using the Internet (% of population)**: measures the share of a country’s population that uses the internet, serving as our grouping variable.
2. **Employment to population ratio, 15+, female (%)**: the modeled ILO estimate of the female employment-to-population ratio for individuals aged 15 and above.
3. **Female share of employment in senior and middle management (%)**: the percentage of senior and middle management positions held by women.

All indicators are measured at the country level for the year 2022.

2.2 Data Cleaning

Data cleaning was performed in Python (see `Cleaning Data.ipynb`). The key steps were:

1. **Filtering for individual countries.** Regional and income-group aggregates (e.g., “Sub-Saharan Africa,” “High income”) were removed using valid ISO 3166-1 alpha-3 country codes from the `pycountry` library. This ensures each observation represents a sovereign nation.
2. **Selecting the analysis year.** We surveyed data availability for each year from 2015 to 2025 across all three indicators. Year 2022 yielded the highest count of countries with complete data across all three datasets (92 countries), making it the optimal choice.
3. **Selecting appropriate disaggregation levels.** The employment dataset contains multiple disaggregation levels (male/female, modeled/national, age groups). We retained only the “female, Modeled, 15+” rows. The internet dataset was filtered to the “total” disaggregation. The management dataset had a single disaggregation per country-year and required no additional filtering.
4. **Identifying common countries.** We computed the intersection of countries with non-missing, non-zero values across all three datasets for 2022, yielding a final sample of **92 countries**.
5. **Handling missing values.** The `Value` column was cast to numeric, and any remaining NaN entries were dropped. No empty strings or additional missing data were found.
6. **Reshaping.** The three filtered datasets were concatenated and pivoted into a wide-format table with one row per country and three columns: internet use, female employment, and female share of senior/middle management.

3 Methodology

3.1 Group Classification

Countries were classified into two groups based on their internet usage rate:

- **High Internet Access:** $\geq 50\%$ of the population using the internet.

- **Low Internet Access:** $< 50\%$ of the population using the internet.

The 50% threshold is consistent with the International Telecommunication Union's (ITU) reported global average internet usage rate for 2019 [2].

For the employment analysis (170 merged countries when considering the broader employment dataset), this yielded 130 countries in the high group and 40 in the low group. For the management analysis (93 countries in the intersection of all three datasets), this yielded 79 countries in the high group and 14 in the low group.

3.2 Normality Assessment

Before selecting our hypothesis testing approach, we assessed normality of each outcome variable within each group using two complementary formal tests and visual diagnostics:

- **Shapiro-Wilk test:** a widely used test for normality based on the correlation between the data and corresponding normal order statistics. The Shapiro-Wilk test is particularly well-suited for our sample sizes ($n = 14$ to $n = 130$), as it maintains good power for small to moderate samples [5].
- **Anderson-Darling test:** a modification of the Kolmogorov-Smirnov test that gives more weight to the tails of the distribution, making it more sensitive to departures from normality in the extremes [1].
- **Visual diagnostics:** histograms and normal Q-Q plots for each group.

We apply both tests because each has different strengths: Shapiro-Wilk is optimal for smaller samples (such as the 14-country low-internet management group), while Anderson-Darling provides additional sensitivity to tail behavior in the larger groups.

3.3 Hypothesis Testing: Mann-Whitney U Test

The Mann-Whitney U test (also called the Wilcoxon rank-sum test) is a nonparametric test for comparing two independent groups. It tests whether one group tends to have larger values than the other by ranking all observations and comparing rank sums. It assumes that the distributions of the two groups that are being compared are the same. We checked this visually by comparing the histograms of the "female employment to population ratio" distribution of the high and low internet groups. We also did the same for the distribution of "female share of senior/mid management" for the high and low internet groups.

For each test, the hypotheses are:

H_0 : The distributions of the two groups are identical (no location shift)

H_a : The distributions differ in location (two-sided)

The test statistic U is computed as:

$$U = R_1 - \frac{n_1(n_1 + 1)}{2}$$

where R_1 is the sum of ranks for group 1 and n_1 is the sample size of group 1.

A **Bonferroni correction** is applied since two tests are conducted on the same sample.

The corrected significance level is $\alpha = 0.05/2 = 0.025$ per test.

4 Results

4.1 Descriptive Statistics

Table 1 and Table 2 present descriptive statistics for the two outcome variables by internet access group.

Table 1: Female Employment Rate (%) by Internet Access Group

Group	n	Mean	Median	SD	Min	Max	IQR
High Internet ($\geq 50\%$)	130	45.43	48.18	14.09	7.86	76.99	15.62
Low Internet ($< 50\%$)	40	53.67	54.07	18.06	3.78	81.05	23.09

Table 2: Female Share of Senior/Middle Management (%) by Internet Access Group

Group	n	Mean	Median	SD	Min	Max	IQR
High Internet ($\geq 50\%$)	79	34.14	35.59	8.98	13.27	59.22	11.32
Low Internet ($< 50\%$)	14	22.13	24.04	8.05	8.87	36.46	11.83

The high internet group shows a lower median female employment rate (48.18%) compared to the low internet group (54.07%), but a substantially higher median female management share (35.59% vs. 24.04%).

4.2 Normality Assessment

4.2.1 Formal Tests

Table 3 summarizes the results of the Shapiro-Wilk and Anderson-Darling normality tests.

Table 3: Normality Test Results ($\alpha = 0.05$)

Variable	Group	Shapiro-Wilk W	A-D A^*2	Decision
Employment	High Internet	0.9507	2.1761	Reject (A-D)
Employment	Low Internet	0.9510	0.5307	Fail to reject
Management	High Internet	0.9732	0.7333	Fail to reject
Management	Low Internet	0.9520	0.3215	Fail to reject

Both the Shapiro-Wilk and Anderson-Darling tests reach consistent conclusions. The Anderson-Darling test clearly rejects normality for the Employment–High Internet group ($A^*2 = 2.17$, $p < 0.001$), and the Shapiro-Wilk test corroborates this with a low W statistic ($W = 0.9507$). For the remaining groups, both tests fail to reject normality; however, this should be interpreted with caution. Notably, the Management–Low Internet group has a small sample size ($n=14$), which limits the statistical power of normality tests to detect true departures from normality. Additionally, cross-national development indicators are broadly known to exhibit skewed distributions, and maintaining methodological consistency across both analyses further supports the use of nonparametric methods. Since at least one major group violates normality, and given these additional considerations, the overall pattern supports the use of nonparametric methods for all group comparisons.

4.2.2 Visual Diagnostics

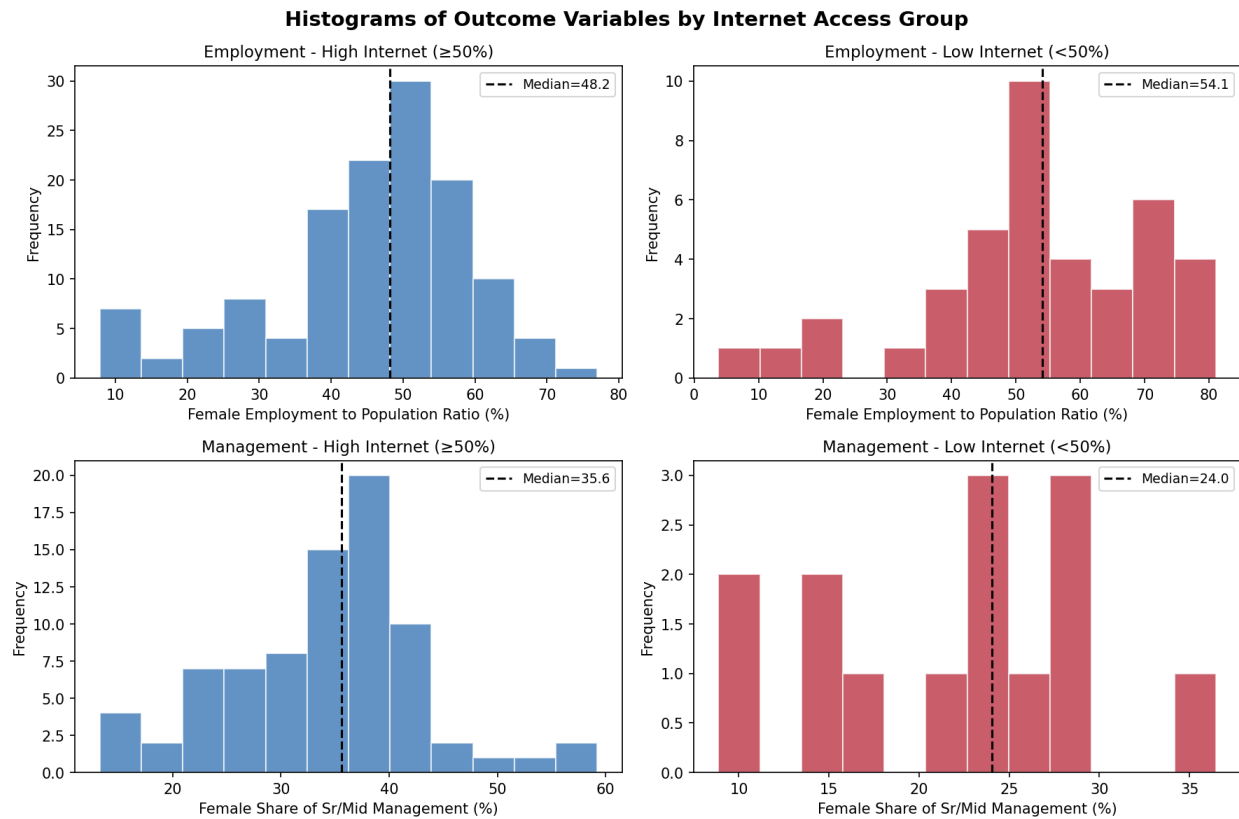


Figure 1: Histograms of outcome variables by internet access group. Dashed lines indicate group medians.

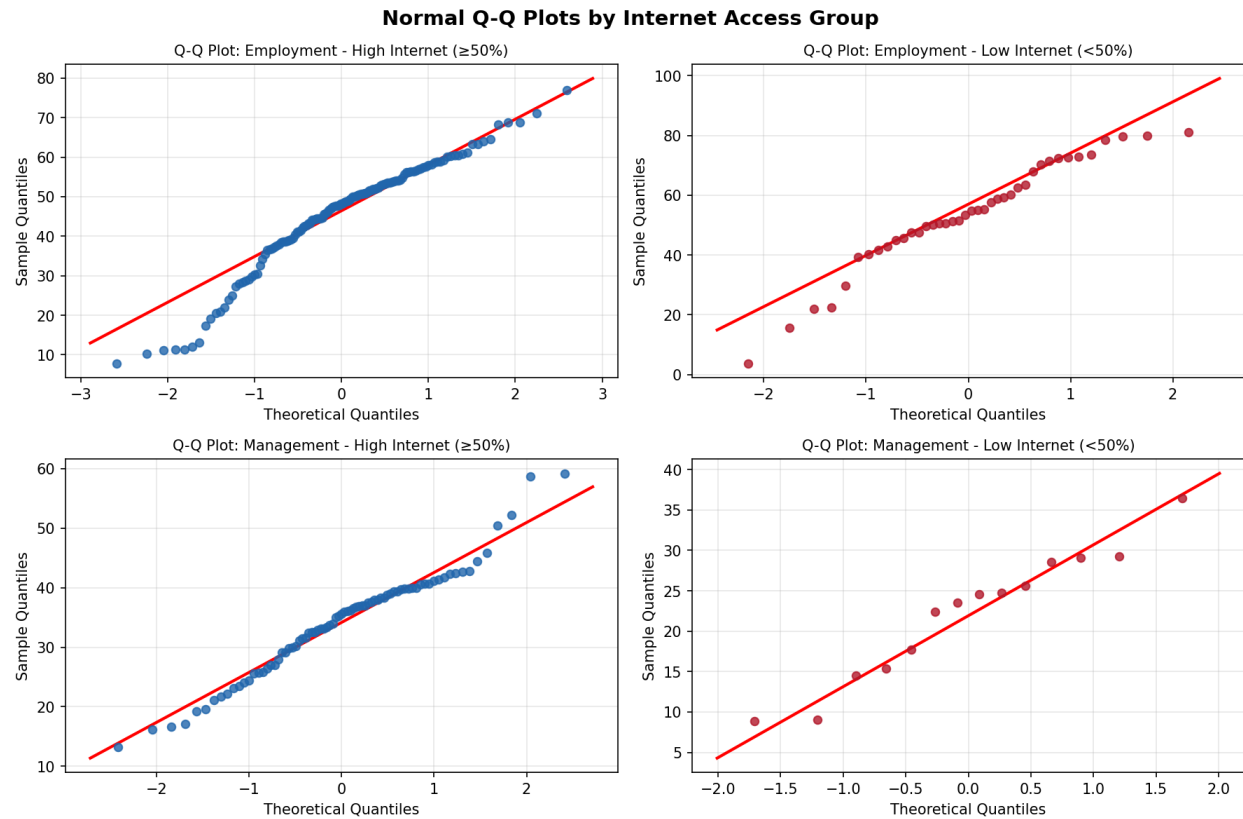


Figure 2: Normal Q-Q plots by internet access group. Departures from the reference line (particularly in the tails of the Employment High Internet group) confirm non-normality and support the use of nonparametric tests.

The histograms (Figure 1) reveal that the employment variable, particularly in the high internet group, exhibits a left-skewed, multimodal distribution. The Q-Q plots (Figure 2) show tail departures in the employment groups. These visual diagnostics corroborate the formal test results and justify the nonparametric approach.

4.3 Mann-Whitney U Test Results

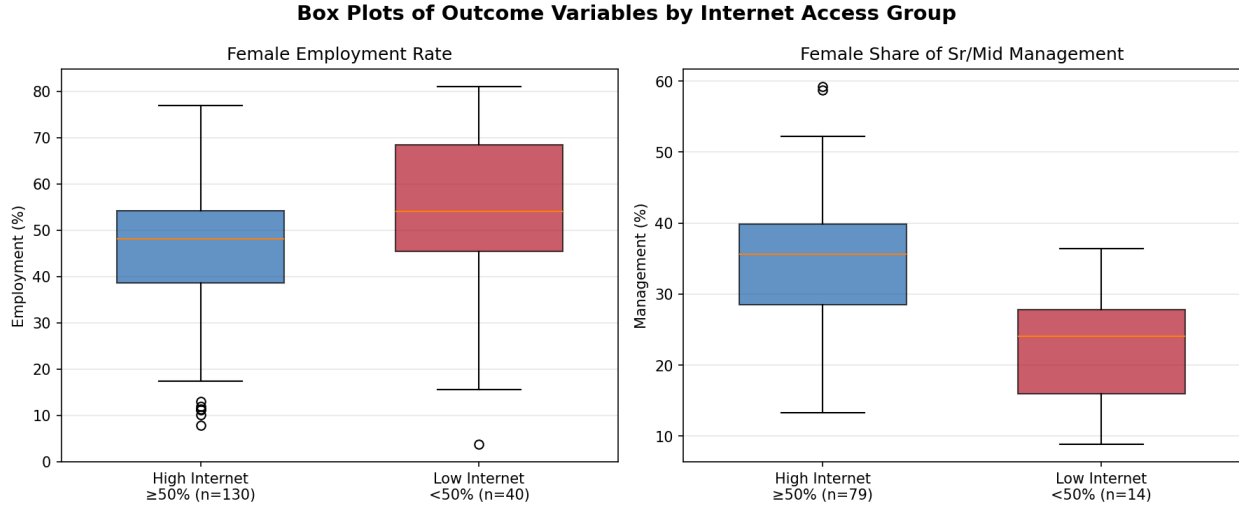


Figure 3: Box plots comparing the distributions of female employment rate and female share of senior/middle management between high and low internet access groups. The management variable shows a clear separation between groups.

4.3.1 Test 1: Female Employment Rate

Table 4: Mann-Whitney U Test, Female Employment Rate

Statistic	Value
n (High Internet)	130
n (Low Internet)	40
W statistic	1798.0
Z -score	-2.9444
p -value	0.0032
Effect size r	0.2258 (small)
Median (High)	48.18%
Median (Low)	54.07%
Decision ($\alpha = 0.025$)	Reject H_0

At the Bonferroni-corrected significance level of $\alpha = 0.025$, we **reject the null hypothesis** ($p = 0.0032$). There is a statistically significant difference in the median female employment rate between the two groups. Notably, the direction is counterintuitive: countries with *low* internet access have a *higher* median female employment rate (54.07%) than those with high access (48.18%). The effect size is small ($r = 0.23$).

4.3.2 Test 2: Female Share of Senior/Middle Management

Table 5: Mann-Whitney U Test, Female Share of Senior/Middle Management

Statistic	Value
n (High Internet)	79
n (Low Internet)	14
W statistic	935.0
Z -score	-4.0987
p -value	0.00004
Effect size r	0.4250 (medium)
Median (High)	35.59%
Median (Low)	24.04%
Decision ($\alpha = 0.025$)	Reject H_0

We strongly **reject the null hypothesis** ($p < 0.0001$). Countries with high internet access have a significantly higher median female share of senior and middle management (35.59%) compared to those with low access (24.04%). The effect size is medium ($r = 0.43$), indicating a substantively meaningful difference.

5 Discussion

Both Mann-Whitney U tests yield statistically significant results at the Bonferroni-corrected $\alpha = 0.025$ level. However, the findings tell different stories about how internet access relates to female economic outcomes.

Employment rate. The result that low-internet countries have higher female employment rates is initially surprising. However, this could be because women’s labor force participation in lower income countries is driven by economic necessity rather than opportunity. This pattern is supported by the strong correlation $r = 0.87$ between country wealth and internet access rates, meaning that low-internet countries are likely also low-income countries [6]. In low-income countries, women are more likely to work in informal sectors, subsistence agriculture, and unpaid family labor, all of which contribute to higher employment-to-population ratios but do not necessarily reflect economic empowerment or quality of employment [8]. Indeed, UN Women (2024) reports that nearly 60% of women’s employment globally is in the informal economy, rising to more than 90% in low-income countries [8]. This suggests that high female employment rates in low-internet countries may reflect survival labor rather than meaningful workforce participation.

Management representation. The strong positive association between internet access and female management share ($r = 0.43$, medium effect) is more aligned with the hypothesis that digital connectivity supports women’s advancement [7]. Countries with higher internet penetration may benefit from greater access to education, remote work opportunities, professional networks, and information about labor rights, all factors that could facilitate women’s entry into leadership positions [4].

5.1 Limitations

Several important limitations should be considered when interpreting these results:

Group size imbalance. The most notable methodological concern is the imbalanced group sizes, particularly for the management analysis where the low-internet group contains only 14 countries compared to 79 in the high-internet group. This imbalance arises naturally from the data: in 2022, the median internet penetration percentage for the countries in the World Bank data is 83.87% and the mean is 77%. The Mann-Whitney U test is robust to unequal group sizes, which supports our use of this test in the analysis. However we do note that specifically, those 14 countries may not be representative of all low-internet nations, as many countries with low internet access simply lack the institutional capacity to collect and report gender-disaggregated management data. Therefore, **we cannot generalize the management finding to all countries with low internet usage**, only to those that report this indicator.

Cross-sectional design. This analysis is cross-sectional and correlational; no causal claims can be made. Internet access and female economic outcomes are both associated with broader national development, and the observed relationships may be driven by confounding factors such as GDP per capita, cultural and religious norms, existing labor market policies, urbanization rates, and educational attainment.

Data availability bias. The 92-country sample is not a random draw from the world’s nations. Countries that participate in international data collection may have stronger statistical infrastructure. This selection effect may bias the sample toward countries where both internet access and female economic participation are higher than the global average.

Measurement limitations. The employment-to-population ratio does not distinguish between formal and informal employment, full-time and part-time work, or paid and unpaid labor. Similarly, “senior and middle management” definitions may vary across countries’ national statistical offices.

Future work could employ regression-based approaches with control variables, use longitudinal panel data to track changes within countries over time, or adopt instrumental variable strategies to better isolate the causal role of internet access.

6 Conclusion

This study finds statistically significant differences in both female employment rates and female management representation between countries with high ($\geq 50\%$) and low ($< 50\%$) internet usage in 2019. The management result in particular provides compelling evidence that internet access is positively associated with women's advancement into leadership roles, with a medium effect size ($r = 0.43$) suggesting a meaningful, not merely statistically detectable, relationship. The employment result, while statistically significant, points in the opposite direction (low-internet countries exhibit higher female employment rates) which upon closer examination reflects the prevalence of informal and agricultural labor in lower-income countries rather than genuine economic empowerment. Taken together, these two findings underscore an important distinction: internet access may matter less for whether women work, and more for the quality and seniority of the work they do.

These findings have practical implications for development initiatives and gender equity policy. Investments in internet infrastructure alone may not directly increase women's aggregate labor force participation, but they appear to support women's upward mobility into higher-quality, decision-making positions. This aligns with broader literature suggesting that digital connectivity offers 'leapfrog' opportunities for women by expanding income-earning potential, employment opportunities, and access to knowledge that can support advancement in formal labor markets [4]. Policymakers should therefore consider complementary interventions (such as digital literacy programs, remote work policies, and targeted leadership training) to maximize the gender equity benefits of expanding internet access, particularly in middle-income countries where internet penetration is growing but women remain under-represented in management.

This study is not without limitations. The cross-sectional and correlational nature of the analysis means no causal claims can be made, and the sample of 63 countries with complete data across all three indicators may not be fully representative of global patterns. Confounding factors such as GDP per capita, regional cultural norms, legal protections for women in the workplace, and existing educational attainment levels may all independently drive both internet access and female economic outcomes. Future research should examine these relationships longitudinally, across multiple years, to assess whether expansions in in-

ternet access within a given country over time are associated with improvements in female management representation, a design that would provide stronger causal evidence than the cross-national comparison employed here.

References

- [1] Anderson, T. W., & Darling, D. A. (1954). A test of goodness of fit. *Journal of the American Statistical Association*, 49(268), 765–769.
- [2] International Telecommunication Union. (2019). *Measuring digital development: Facts and figures 2019*. <https://www.itu.int/en/ITU-D/Statistics/Documents/facts/FactsFigures2019.pdf>
- [3] Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1), 50–60.
- [4] OECD. (2018). *Bridging the digital gender divide: Include, up-skill, innovate*. <https://www.valored.it/wp-content/uploads/2018/11/bridging-the-digital-gender-divide-key-messages-ridotto-pages-1-12-ilovepdf-compress.pdf>
- [5] Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3/4), 591–611.
- [6] Silver, L., Smith, A., Johnson, C., Taylor, K., Jiang, J., Anderson, M., & Rainie, L. (2016, February 22). *Internet access growing worldwide but remains higher in advanced economies*. Pew Research Center. <https://www.pewresearch.org/global/2016/02/22/internet-access-growing-worldwide-but-remains-higher-in-advanced-economies/>
- [7] UN Women. (2023). *Innovation and technological change, and education in the digital age for achieving gender equality and the empowerment of all women and girls: Report of the Expert Group Meeting*. <https://www.unwomen.org/sites/default/files/2023-02/CSW67-Expert-Group-Meeting-report-en.pdf>
- [8] UN Women. (2024). *Facts and figures: Economic empowerment*. <https://www.unwomen.org/en/what-we-do/economic-empowerment/facts-and-figures>
- [9] Verick, S. (2014). Female labor force participation in developing countries. *IZA World of Labor*, 87. <https://doi.org/10.15185/izawol.87>

- [10] Viollaz, M., & Winkler, H. (2020). Does the internet promote gender equality in the labor market? Evidence from low- and middle-income countries. *World Bank Research Observer*, 37(1). <https://openknowledge.worldbank.org/server/api/core/bitstreams/8f79abb6-1407-54ac-a614-cd06097c9480/content>
- [11] World Bank. (2022). *Gender Data Portal: Indicators*. <https://genderdata.worldbank.org/en/indicators>